ACS Sustainable Chemistry & Engineering

Research Article

Cite This: ACS Sustainable Chem. Eng. 2019, 7, 2998−3009

pubs.acs.org/journal/ascecg

Preparation of High-Strength Sustainable Lignocellulose Gels and Their Applications for Antiultraviolet Weathering and Dye Removal

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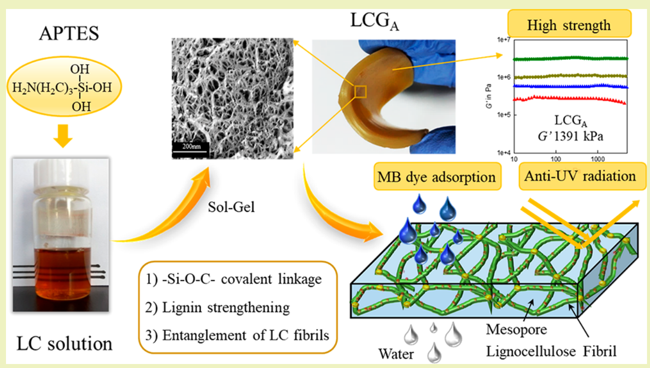
Supporting Information

ABSTRACT: The synthesis of functional lignocellulose-based gels from sustainable biomass has received considerable attention in material chemistry. In this study, robust, porous, and lignin-containing lignocellulose hydrogels were prepared based on the sol−gel process. The lignin-containing lignocellulose materials were dissolved (or dispersed) in *N*-methylmorpholine-*N*-oxide (NMMO) and cross-linked with a silane coupling agent 3-aminopropyltriethoxysilane (APTES), followed by coagulation and solvent exchange to form gel structures. The formation of Si−O−C cross-links among the lignocellulosic fibrils as a result of the addition of APTES was mainly responsible for the strength improvement. Furthermore, the presence of lignin also contributed to the enhancement of strength properties. The as-prepared lignocellulose gels were further characterized using rheological and compression tests. The dynamic storage modulus of APTES-reinforced lignocellulose gel (LCG_A) can reach up to 1391 kPa and the compressive modulus up to 96 kPa, which shows a 3-fold increase in viscoelastic properties and a 2-fold increase in compressive strength compared with unmodified gel. In addition, the obtained LCG_A gel exhibits unique properties, such as antiultraviolet weathering and dye removal via adsorption, and their potential applications were explored.



KEYWORDS: *Lignocellulose gel, Silane coupling agent, Lignin, Reinforcement, Antiultraviolet, Dye removal*

■ INTRODUCTION

Hydrogels are 3D polymer networks through cross-linking with chemical or physical bonding, which can adsorb large amounts of water.[1] Due to their biocompatibility, and biodegradability, cellulose-based gels have received much attention and have applications in tissue engineering,[2] blood purification,[3] sensors,[4] water purification,[5] and thermal superinsulation,[6] as well as biomedicine and cosmetics.[7,8]

Lignocellulose, consisting of cellulose, hemicellulose, and lignin, is also an excellent candidate for gel fabrication.[9,10] The lignocellulose-based hydrogels are mainly prepared using sol−gel technology.[9] The solvent systems for lignocellulose include: dimethyl sulfoxide/tetrabutyl ammonium fluoride,[11] dimethyl sulfoxide/lithium chloride,[12] dimethyl sulfoxide/*N*-methylimidazole,[13] and ionic liquid (IL) solvent systems.[14] NMMO/H_2O was also used to prepare a lignocellulose solution under moderate conditions.[15] The obtained lignocellulose solution can be used for the synthesis of high-performance lignocellulose gel products.

Many of these lignocellulose hydrogels are fragile and brittle,[16] limiting their practical applications. The application of such hydrogels, for example in wastewater remedy by adsorption, requires strong mechanical properties.[17] Numerous studies have made efforts to enhance the mechanical properties of lignocellulose hydrogel, including nanocomposite gels,[18] double-network gels,[19] and topological gels.[20] Some reports utilize the physical and chemical cross-linking to strengthen the mechanical properties of gels.[17]

Silane- based chemistry has been successfully applied to cellulose composites, so that effective cross-linking is formed in the cellulose networks.[21] Matuana et al. demonstrated that the cross-linking of silane coupling agents can reinforce mechanical performance of the PVC/cellulosic fiber composites, and their tensile strength can increase up to 36%.[22] It was reported by Park et al. that the silanes can be used to improve the interfacial compatibility of jute fibers and polypropylene (PP), enhancing their strength.[23] The reactive silanol groups of the silane coupling agent after hydrolysis can form −Si−O−C− bonds with hydroxyl groups of cellulose, enhancing the interfibril networks.[21] In the present study, the silane-based

Received: September 3, 2018
Revised: November 7, 2018
Published: January 2, 2019

Table 1. Compositions of the Structural Components of the Cellulose and Lignocellulose Samples

samples	carbohydrate			lignin			ash	DP
	glucan	xylan	total	KL	ASL	total		
cellulose	74.2 ± 0.5	24.6 ± 0.2	98.8 ± 0.7	0.3 ± 0.1	0.2 ± 0.0	0.5 ± 0.1	0.0 ± 0.0	736
lignocellulose	69.8 ± 0.5	21.3 ± 0.4	91.1 ± 1.0	6.2 ± 0.5	0.3 ± 0.0	6.5 ± 0.5	0.2 ± 0.1	815

cross-linking chemistry was followed, producing lignocellulose hydrogel with exceptionally high strength properties (storage modulus, compressive strength).

It is well-known that lignin in natural woody plants delivers their required strength. Lignin can also be considered as the essential glue between cellulose and hemicellulose, providing mechanical strength for wood and rigidity to resist external forces.[24] There are many publications in the literature[25,26] documenting the relationship between the fiber lignification process (increasing the lignin content) and its strength development, demonstrating the critical importance of lignin to the strength of native woody tissues. Lignification/lignin biosynthesis is a complex process,[27] which starts with the deamination of the aromatic amino acid phenylalanine, followed by a series of hydroxylation, methylation, and reduction resulting in the production of the basic units of the lignin complex.[28] With an increase of lignification degree, the lignin polymer can provide mechanical strength and rigidity for secondary cell walls to allow plants to grow tall, thereby enhancing the strength of the plant.[25] On the other hand, some research found that dissolved lignins tend to associate with one another to form lignin nanoparticles in both organic and aqueous media.[29−31] Inspired by nanocomposites, lignin nanoparticles can be used for enhancement of mechanical strength of the lignocellulose gel.

The presence of pollutants such as organic dye in water is a serious concern because they have a negative impact on the environment and their toxicity damages human body.[32] The sustainable biobased hydrogel could be utilized as a highly efficient adsorbent for wastewater treatment, due to its porous network structure. Moreover, compared with conventional adsorbents (such as activated carbon), the biobased hydrogel has an advantage of easy separation to avoid producing toxic chemical sludge. Zhou et al. reported a cellulose-based hydrogel was prepared for the methylene blue (MB) dye removal from aqueous solution.[33] Yu et al. prepared a lignin-based hydrogel for remediation of cationic dye-contaminated effluent, and the hydrogel possesses superabsorbent capacity for MB.[34]

In this study, novel lignin-containing lignocellulose gels with porous structure and exceptionally high strength were prepared, which were based on the utilization of lignocellulose raw materials from poplar, through the sol−gel process. A silanol precousor, 3-aminopropyltriethoxysilane (APTES), was used, and its hydrolysis leads to the formation of silane triols that are effective coupling agents to enhance the mechanical strength of the lignocellulose hydrogel. Biomass-based hydrogels have been reported.[16,35,36] The application of such hydrogels, for example, in wastewater remedy by adsorption, requires strong mechanical properties.[17] In our approach, lignin-containing poplar pulp was dissolved (or dispersed) in an NMMO monohydrate solution. APTES was also dissolved in NMMO/H_2O. The lignocellulose gels were subsequently formed by mixing these two solutions and using ethanol for solvent exchange. The Si−O−C covalent bonds were formed in situ, which were responsible for the enhanced mechanical strength of lignocellulose hydrogels. The obtained lignocellulose gels were measured using scanning electron microscopy (SEM), confocal laser scanning microscopy (CLSM), N_2 adsorption−desorption measurements, X-ray diffraction (XRD), Fourier transform infrared (FTIR), and rheological and compression tests. In addition, the as-prepared lignocellulose gels can present other interesting properties, such as antioxidant properties and good adsorption capacity. Accordingly, such a lignin-containing gel can be of great potential to serve as high value-added products for outdoor and environmental applications.

EXPERIMENTAL SECTION

Materials. Lignin-containing lignocellulose (LC) (6.2 ± 0.5% Klason lignin and 0.3 ± 0.0% acid soluble lignin) was prepared from poplar wood chips via the kraft pulping process. The lignin and sugar contents of samples were measured by following the NREL protocol.[37] The Klason lignin (KL) content was determined after a 2-step sulfuric acid hydrolysis to remove the carbohydrates: 72% strong sulfuric acid and 3% dilute sulfuric acid; the residual solids were filtered, collected, dried, and weighted, which were known as the Klason lignin, while the filtrate sample separated from the 3% dilute sulfuric acid hydrolysis was determined using an UV−vis spectrometer (UV-1800, Shimadzu) at 205 nm wavelength as the acid-soluble lignin. A calibration curve was then obtained.[38] The ash content was determined as the residue by combustion at 575 °C.[39] The sugars were quantified by using high performance liquid chromatography (HPLC, Agilent 1200 Series, Santa Clara, CA), equipped with a refractive index detector (RID). The DP of lignocellulose sample (lignin content: 6.5%) was measured by following a literature procedure.[40] Due to the presence of lignin, a bleaching step was carried out. The obtained dry bleached sample was dissolved in 0.5 M copper ethylenediamine solution (30 mL). The intrinsic viscosity of the cellulose solution was measured using an Ubbelohde viscometer at 20 °C. The viscosity-average degree of polymerization (DP) of the cellulose sample was calculated from the intrinsic viscosities using the Mark−Houwink−Sakurada equation.

Lignin-free cellulose (0.2 ± 0.05% Klason lignin and 0.3 ± 0.0% acid-soluble lignin) acted as a control sample for comparison with LC. The cellulose sample was obtained by ECF bleaching sequence of lignocellulose materials.[41] Briefly, the lignocellulose pulp (30 g oven dry, 91.2 g wet weight) was subjected to a three-stage bleaching sequence consisting of DED to remove lignin, with the following conditions:

- D_1: 1% ClO_2, 75 °C, 30 min;
- E: 3% sodium hydroxide, 75 °C, 45 min;
- D_2: 2% ClO_2, 75 °C, 60 min.

The lignin-free sample was dissolved in copper ethylenediamine solution for the measurement of its DP. The physical characteristics of dried cellulose and lignocellulose samples were shown in Table 1.

The 50 wt % *N*-methylmorpholine-*N*-oxide (NMMO) aqueous solution was purchased from Shanghai Aladdin Chemical Co. Ind., China. 3-Aminopropyltriethoxysilane (APTES) was obtained from Nanjing Chen Gong Organic Silicon Material Co. Ltd. All other reagents were purchased from Nanjing Chemical Reagent Co. Ltd. The 50 wt % NMMO aqueous solution was concentrated to 86.7 wt % NMMO aqueous solution (NMMO/H_2O) as described by Rabideau et al.[42]

Preparation and Cross-Linking of Lignocellulose Gels. The lignocellulose gels were prepared by regeneration and gelation of

Figure 1. Schematic for the preparation of APTES-reinforced lignocellulose gel (LCG_A) and mechanical enhancement.

lignocellulose solution, as shown in Figure 1. The cellulose sample (largely lignin-free) and lignocellulose sample (lignin content of 6.5%) were dissolved (or dispersed) in $NMMO/H_2O$ via a two-step pretreatment (glycerol swelling and mechanical extrusion) to give 3% clear solutions, as described in our previous study.[15] Glycerol can effectively swell the cellulose or lignocellulose samples, resulting in less compact cellulose or lignocellulose structures. The mechanical extrusion, a lab-scale screw extrusion device,[43] is to disrupt the cellulose/lignocellulose fibers, which will facilitate the subsequent cellulose dissolution process in $NMMO/H_2O$. Briefly, 3 g (oven dry) of lignocellulose (or cellulose) samples was soaked in 24 g of glycerol for 2 h at 120 °C. The glycerol-swelled samples were then mechanically extruded in the lab-scale screw extrusion device. After pretreatment, the samples were thoroughly washed by water and subsequently freeze-dried. The obtained samples (0.9 g, oven dry) were dissolved (or dispersed) in 29.1 g of $NMMO/H_2O$ solution at 85 °C for 3 h, thereby obtaining a clear lignocellulose $NMMO/H_2O$ dispersion (or cellulose $NMMO/H_2O$ solution).

The homogeneous lignocellulose $NMMO/H_2O$ dispersion (or cellulose $NMMO/H_2O$ solution) was poured into a glass dish and then transferred to a sealed vessel containing an appropriate ethanol coagulation bath for regeneration. After gelation, the residual NMMO solvent in lignocellulose (or cellulose) gel was then displaced with water by adding an excessive amount of distilled water in a beaker, which was on a lab shaker for 3 h. The process was repeated until the pH of the solution reached neutral, indicating that the residual NMMO is washed away. The washed solid cakes were stored in distilled water to obtain the cellulose gel (CG) and lignocellulose gel (LCG).

In the present work, a mixture of $NMMO/H_2O$ was used, which contains 13.3% water. An amount of 1 wt % APTES in $NMMO/H_2O$ was prepared by using 0.1 g of APTES and 10 g of $NMMO/H_2O$, and a homogeneous APTES in $NMMO/H_2O$ was obtained. APTES undergoes hydrolysis, resulting in the formation of silane triols (Figure S1).[21] Subsequently, various amounts of already-prepared APTES/$NMMO/H_2O$ solution were added to the desired amounts of LC solution and constantly stirred for 1 h, and the modified LC solution was obtained. For example, 0.9 g of 1 wt % APTES in $NMMO/H_2O$ solution was added to 30 g of 3 wt % lignocellulose in $NMMO/H_2O$ solution to react at 100 °C for 1 h, thereby obtaining the APTES-modified LC solution. In this system, the addition amount of APTES was 1 wt % on lignocellulose material. After that, the APTES-modified lignocellulose dispersion (or cellulose solution) was regenerated and gelated with ethanol coagulation bath.[42] The regenerated APTES-modified lignocellulose (or cellulose) gel samples were thoroughly washed by following the same procedures detailed above, until reaching a neutral pH of the washing solution. Finally, the APTES-modified cellulose gel and lignocellulose gel were obtained and abbreviated to CG_A and LCG_A, respectively.

Structural Characterizations. TEM analyses: Using a homogenizer, the LCG_A samples were broken into small particles of about 20 μm, which were then transferred to carbon-coated copper grids and examined on a JEOL 2010 transmission electron microscope with 20 kV acceleration voltages.

SEM analyses: All gel samples (CG, LCG, CG_A, and LCG_A) were immersed into liquid nitrogen and subjected to freeze-drying in a vacuum freeze-dryer (LABCONCO) at −50 °C for 3 days.[18] The freeze-dried lignocellulose gel samples were carbon coated and then examined on a JEOL-JSM 7600F scanning electron microscope (SEM) (JEOL, Japan) operated at a 15 kV acceleration voltage.

N_2 adsorption−desorption analyses: The freeze-dried lignocellulose gel samples were performed at 77 K using a V-Sorb 2800P (Gold APP Instruments Cooperation, China) to determine pore properties of lignocellulose aerogels. The moisture was removed from the samples by degassing at 100 °C for 8 h. Brunauer−Emmet−Teller (BET) analyses were used to determine the surface area through calculating the adsorbing amount of N_2 at various relative vapor pressures. Meanwhile, the pore-size distribution of hydrogels was obtained using the Barrett−Joyner−Halenda (BJH) method.

Confocal Laser Scanning Microscopy (CLSM). The acridine orange (AO) solution of 10^{-6} M was obtained by dissolving a specified amount of AO in deionized water. The lignocellulose hydrogel (0.2 mg (oven dry)) was stained with 20 mL of fresh 10^{-6} M AO solution in a 25 mL vial. The hydrogel was subjected to stirring for 1 h under the avoiding light conditions. After labeling, the deionized water was used to remove the residual AO dye.

The images of labeled hydrogels were observed on a confocal laser scanning microscope (Leica DM IRE2) with illumination light of 488 nm. Subsequently, the image of lignin fluorescence in labeled hydrogels was able to be collected at wavelengths between 515 and 540 nm.

The Fourier transform infrared (FT-IR) spectra of lignocellulose aerogels were obtained during the range of 4000−400 cm^{-1} on a VERTEX 80 V spectrometer (Bruker, Germany). X-ray diffractions (XRD) of original LC materials and freeze-dried LCG_A were performed on an X-ray diffractometer (Ultima-IV). The X-ray source was a sealed, 2.2 kW Cu X-ray tube, maintained at an operating current of 40 kV and 25 mA. The wavelength was 1.54 Å, and the diffraction angle 2θ was from 4° to 50°.

Mechanical Properties. Rheological tests of hydrogels were measured using a RS6000 Rheometer (HAAKER, German). A P20 TiL Platte and a P20 TiL cone plate could be used. All specimens

DOI: 10.1021/acssuschemeng.8b04413
ACS Sustainable Chem. Eng. 2019, 7, 2998−3009

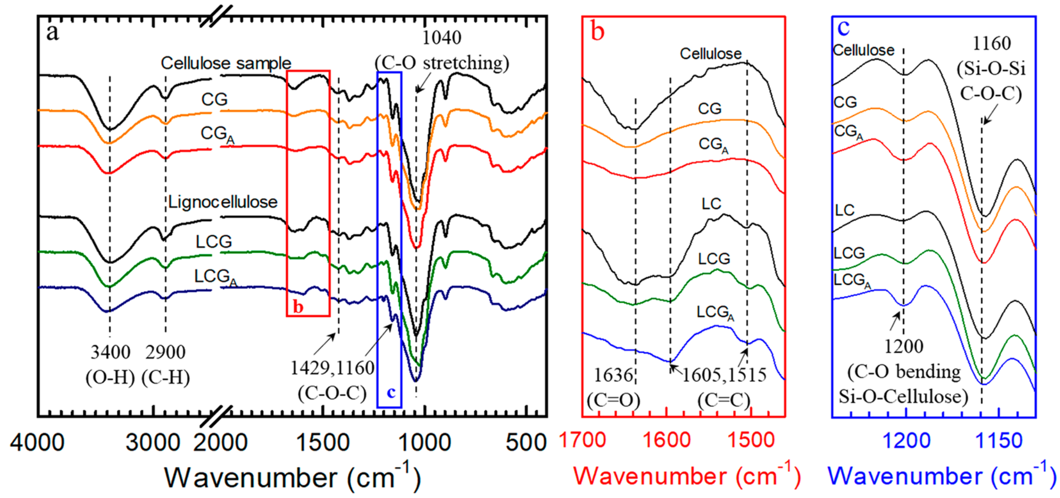


Figure 2. (a) FTIR of original cellulose sample, lignocellulose (LC), cellulose gel (CG), lignocellulose gel (LCG), APTES-reinforced cellulose gel (CG_A), and lignocellulose gel (LCG_A). (b) and (c) Part FTIR of all samples from (a).

(CG, CG_A, LCG, and LCG_A) were prepared as 2 mm thick disks with diameters equal to the P20 TiL plate. The specimens were subjected to a stress−sweep test at an ambient temperature. Storage modulus (G') and loss modulus (G'') were recorded for lignocellulose gels. The samples were measured three times.

The compression performances of lignocellulose hydrogels were analyzed using an SLBL-500N tensile-compressive tester (SHIMADZU, Japan). The circular specimens with 25 mm in diameter were prepared, and thickness was measured. Various lignocellulose hydrogels were measured at least three times. The compressive modulus (Es) was the slope of the stress−strain curve in the elastic region.

Accelerated Weathering Test. The freeze-dried CG and LCG_A were placed in an UV radiation tester (BZS250GF-TS, China) equipped with UV irradiance (UVS250GK2-1I) for 72 h.[44] The UV irradiance was 250 W/m^2 at 254 and 185 nm wavelength with photon energy of 472 and 647 kJ/mol, respectively. The weight change of the tested samples was evaluated after duration of 5 min, 30 min, 1 h, 8 h, 16 h, 32 h, 48 h, and 72 h.

Dye Adsorption Studies. The adsorption capacities of CG and LCG_A were measured by performing experiments using methylene blue (MB). The 25 mg adsorbents (oven dry) were placed in the 250 mL conical flasks with 100 mL of MB solution (50 mg/L).[45] The mixtures were shaken for 24 h using a thermostatic shaker (MaxQ4000, Thermo scientific, America) at 30 °C and 100 rpm. The amounts of MB adsorbed on hydrogels were obtained by determining the concentrations of MB solution using an UV−vis spectrophotometer (UV-1800, Shimadzu) at 663 nm as λ_{max} of MB. Equation 1 below was used to calculate the amounts of MB adsorbed on hydrogel

$$Q_e = \frac{V(C_0 - C_e)}{m} \tag{1}$$

where C_0 and C_e are the initial and equilibrium concentrations of MB (mg/L), respectively. V represents the volume of solution (mL), and m is the oven dry mass of prepared hydrogel (g).

The pseudo-first-order and pseudo-second-order kinetic models were used to estimate adsorption kinetics of hydrogels and were determined in accordance with eq 2 and eq 3 below

$$\log(Q_e - Q_t) = \log Q_e - \frac{k_1 t}{2.303} \tag{2}$$

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{t}{Q_e} \tag{3}$$

where Q_e and Q_t are the adsorption capacities of MB (mg/g) at equilibrium time and time t, and K_1 (min^{-1}) and K_2 (g/(mg min)) are the adsorption rate constants of the pseudo-first-order and pseudo-second-order adsorption models, respectively.

RESULTS AND DISCUSSION

Preparation of APTES-Reinforced Lignocellulose Gels (LCG_A) and Their Mechanical Enhancement. The lignocellulose (LC) was dissolved (or dispersed) in the NMMO/H_2O solvent (Figure 1a). APTES was also dissolved in NMMO/H_2O solution, and the formed silanols from the APTES hydrolysis (Figures 1b and S1)[21] are reactive in the cross-linking reactions. The hydrolyzed APTES (dissolved in NMMO/H_2O) was added into the lignocellulose NMMO/H_2O solution. During the LCG_A formation process, −Si−O−C− covalent linkages are formed between cellulose and the hydrolyzed APTES, through the reaction of hydroxyl groups and silanol groups under the heating conditions (100 °C, 1 h).

The reaction mechanism is presented in Figures 1b and S1. After hydrolysis of APTES, silanols first form hydrogen bonding with hydroxyl groups of cellulose.[46] Under heating conditions, covalent −Si−O−C− bonds are formed.[47] Thus, in this work, the hydrolyzed APTES are effective cross-linkers of the regenerated lignocellulose/cellulose. The direct evidence on the covalent linkages of APTES-modified lignocellulose (or cellulose) is provided by FTIR (Figure 2). The resultant covalent linkages strengthened the mechanical properties of lignocellulose gels.

Furthermore, the presence of lignin is partially responsible for the excellent mechanical strength of the lignocellulose gel, which can be confirmed by improvement of mechanical strength of LCG compared with CG (results in next section). During the coagulation process of LCG_A, lignin molecules can associate with each other in ethanol to form a "nanoparticle analogue",[29] enabling the lignin to be used as a rigid phase precipitated in the lignocellulose gel network. In addition, the entanglements between the lignocellulose fibrils will contribute to the strength properties because of the formation of enhanced intermolecular hydrogen bonding between the regenerated cellulose in the CG or LCG samples (Figure 2).

As shown in Figure 2, FTIR spectra of the original cellulose sample, lignocellulose sample, CG, LCG, APTES-reinforced CG_A, and LCG_A, all show obvious adsorption peaks between

DOI: 10.1021/acssuschemeng.8b04413
ACS Sustainable Chem. Eng. 2019, 7, 2998−3009

3300 and 3450 cm^{-1}, which represent O−H stretching of these materials. Compared with the original materials (cellulose sample and lignocellulose sample of Figure 2), the intensity of O−H stretching vibrations of the CG and LCG samples decreases, which is caused by the enhanced hydrogen bonding of the regenerated cellulose/lignocellulose gels.[9] The other adsorption peaks at 2900 cm^{-1}, 1429/1160 cm^{-1}, and 1040 cm^{-1} appear in all of the hydrogels, which are attributed to vibrations of the C−H stretching, the symmetric and antisymmetric stretching vibration of the C−O−C bond, and C−O stretching from cellulose.[48]

Compared with lignin-free cellulose and CG samples (Figure 2b), the lignin-containing lignocellulose and LCG samples have unique adsorption peaks at 1636 cm^{-1} and 1605 cm^{-1}/1515 cm^{-1}, which represent carbonyl (C═O) and aromatic phenyl (C═C) of lignin.[49] These results support the presence of lignin in the lignocellulose and LCG samples.

As shown in Figure 2, the intensities of O−H stretching vibrations (3300−3450 cm^{-1}) from CG_A and LCG_A decrease, which is attributed to the condensation reaction between the hydroxyl group and the silanol group.[50] The peak at 1200 cm^{-1} is attributed to both C−O bending vibration[51] and Si−O−C stretching vibrations. The peak at 1200 cm^{-1} increases (Figure 2c), confirming the formation of Si−O−cellulose linkages through condensation reactions from the hydroxyl groups of cellulose and silanol groups of cross-linking agents. In another study, the characteristic vibration of Si−O−C linkages around 1200 cm^{-1} was reported by Panaitescu et al., who studied the effect of organosilane treatment on the surface properties of hemp fibers.[52] The peak at 1160 cm^{-1} is attributed to both C−O−C and Si−O−Si vibrations (their overlapping at about 1160 cm^{-1}).[52]

Figure S2 shows the FTIR spectra of APTES-reinforced lignocellulose hydrogel with APTES concentration of 1 and 1.5 wt %. The peak intensity at 1160 cm^{-1} (Si−O−Si, C−O−C) of 1.5 wt % APTES-reinforced lignocellulose hydrogel (LCG_A-1.5) is higher than that of 1 wt % APTES-reinforced lignocellulose hydrogel (LCG_A-1.0). The occurrence of more Si−O−Si bonds for the LCG_A-1.5 sample may suggest that more silanol groups of the hydrolyzed APTES would undergo self-condensation when the APTES dosage exceeded 1%.

Morphological and Structural Properties of Lignocellulose Gels. The XRD spectra of the original lignocellulose (LC) and the prepared LCG_A are shown in Figure 3a. It is evident that the XRD patterns of LC raw material are different from those of regenerated LCG_A. The LC raw material has the typical cellulose I structure, with the peaks at $2\theta = 14.7°$, 16.8°, and 22.7°, corresponding to the diffraction planes of $1\bar{1}0$, 110, and 200, respectively.[53] On the other hand, the LCG_A sample (made from the LC sample, which was first dissolved in the NMMO/H_2O solution, followed by the cellulose regeneration process), displays a wide peak at 20.4°, attributed to the 110 crystallographic planes,[54] suggesting the conversion of cellulose I to cellulose II in the NMMO/H_2O dissolution process and regeneration process, which is consistent with that in the literature.[55] NMMO is a powerful solvent that can overcome the inter- and intramolecular hydrogen bonds in cellulose, resulting in a homogeneous cellulose NMMO solution.[56] The hydrogen bond network is reconstructed during the cellulose regeneration/precipitation process.[57]

The morphology and porous structure of LCG_A is observed through TEM and SEM, as shown in Figure 3b and 3c. The 3D mesoporous network structure of LCG_A can be observed from

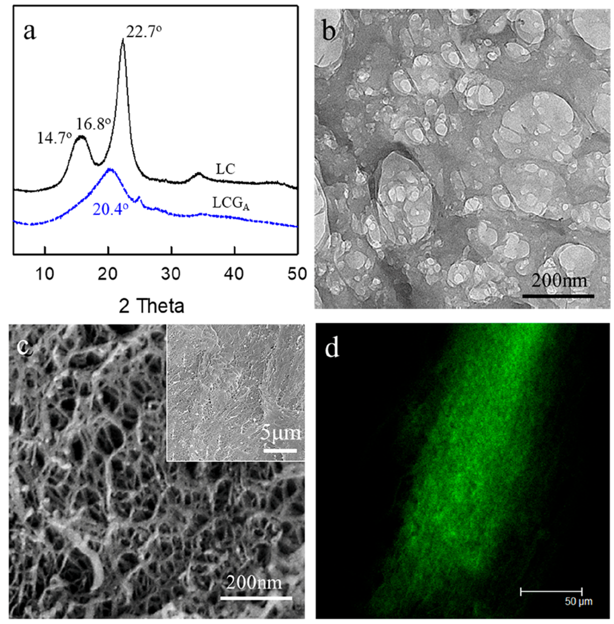


Figure 3. (a) XRD patterns of original lignocellulose (LC) and the APTES-reinforced lignocellulose gel (LCG_A). (b) TEM image of lignocellulose fibril network of LCG_A. (c) SEM image of LCG_A. (d) CLSM characterization confocal image of LCG_A labeled with AO.

TEM and SEM images, which is assembled by nanosized lignocellulose fibrils. A micron-sized SEM image also shows the compact and mesoporous internal structure of LCG_A.

The confocal images of LCG_A labeled with AO are shown in Figure 3d. The LCG_A sample shows strong fluorescence due to the existence of lignin.[58] In contrast, with the absence of lignin, the cellulose gel (made of pure cellulose) is fluorescent free (Figure S3). As shown in Figure 3d, lignin is well distributed in LCG_A, indicating that three components (cellulose, lignin, hemicellulose) are simultaneously regenerated to form the lignocellulose fibrils. The association and reabsorption of lignin can occur in the formation process of the lignocellulose fibril network, and thus lignin can be a reinforcing phase due to its rigidity, which provides a positive effect to the mechanical property of lignocellulose gel.[59] The distribution of lignin along the *Z* axis optical section is shown in Figure S4, indicating that lignin is well dispersed in the lignocellulose hydrogel.

The N_2 adsorption−desorption measurements were used to further analyze the pore structure of the hydrogels, as shown in Figure 4. The absorption−desorption isotherms of CG, LCG, and LCG_A exhibit the type IV curve (IUPAC classification), and the main adsorption occurs in the region of high relative pressure, indicating that all of the pore diameter of CG, LCG, and LCG_A is greater than 10 nm.[60] Compared with CG, the absorption−desorption isotherms of LCG do not show a significant change, confirming that the presence of lignin has no evident effect on pore structure and internal surface area of the gel. As shown in Table 2, the pore size decreases from 18.5 nm (LCG) to 14.6 nm (LCG_A), while the specific surface areas increase from 113.4 m^2 g^{-1} (LCG) to 167.4 m^2 g^{-1} (LCG_A). The LCG_A exhibits the highest surface area, indicating that the APTES-induced cross-linking in the network leads to an increase in the specific surface area of lignocellulose hydrogel. A high specific surface area is beneficial as biomass adsorbent. In addition, Table 2 further shows that the hydrogels prepared

DOI: 10.1021/acssuschemeng.8b04413
ACS Sustainable Chem. Eng. 2019, 7, 2998−3009

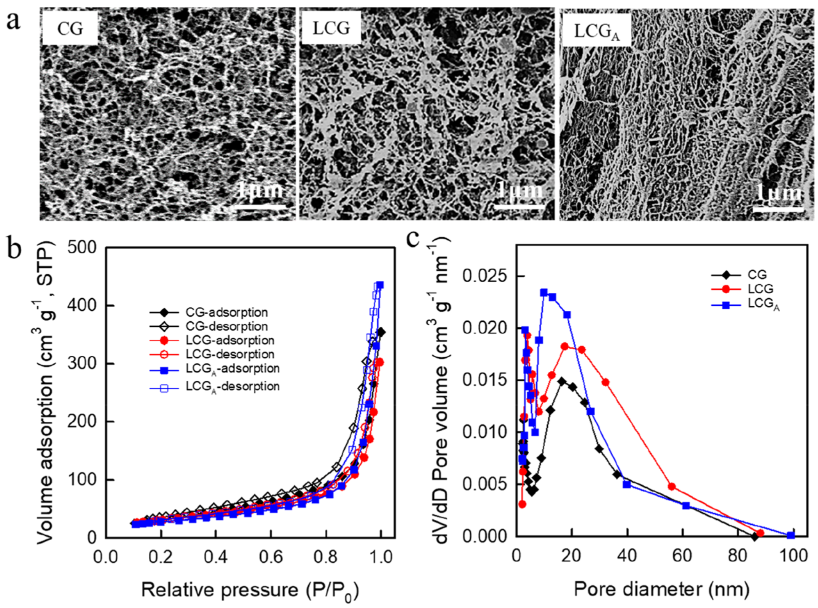


Figure 4. SEM images of nanopore structure of cellulose gel (CG), lignocellulose gel (LCG), and APTES-reinforced lignocellulose gel (LCG_A) (a). N_2 adsorption–desorption isotherms (b) and mesopore size distribution (c) of CG, LCG, and LCG_A.

Table 2. Pore Size and Specific Surface Area of Lignocellulose Hydrogels Prepared in the Present Study and Their Comparison with Those of Literature Work

samples	S_{BET} (m^2/g)	average pore diameter (nm)	ref
CG	120.7	16.3	present work
LCG	113.4	18.5	present work
LCG_A	167.4	14.6	present work
[a]NFT5	80.7	10.5	ref 9
[b]LCNFT-11	102.6	12.3	ref 61

[a]NFT5: Lignocellulose gel prepared by dissolving lignocellulose in ionic liquids with liquid nitrogen freezing–thawing cycle treatment of 5 times. [b]LCNFT-11: Lignocellulose gel with 11 wt % concentration of lignocellulose.

in the present work exhibit higher specific surface area compared with those of the literature, supporting the conclusion that the obtained hydrogels will have enhanced pollutant adsorption.

The internal morphologies of lignocellulose gels are characterized using SEM as shown in Figure 4a, and the corresponding BJH measurement is applied to analyze the pore size distribution of lignocellulose gels (Figure 4c). A broad size distribution is shown in all of the lignocellulosic gels, where the diameter of the pores is approximately 10–25 nm (Table 2). The SEM images of lignocellulosic gels show 3D porous structures formed by randomly oriented fibrils. After cross-linking with APTES, LCG_A demonstrated a dense and compact network structure.

Mechanical Properties of Lignocellulose Gel. The viscoelasticity and compressive properties are shown in Figure 5. The dynamic storage modulus (G') and loss modulus (G'') of the materials represent the stiffness and energy dissipation of hydrogel, respectively.[62] Figure 5a and 5b shows that G' values are much higher than G'' values for all hydrogels, suggesting the hydrogels networks exhibit elastic characteristics.[63]

The G' and G'' values of APTES-reinforced hydrogel (CG_A, LCG_A) were apparently higher than those of unmodified hydrogel (CG, LCG), indicating that the APTES cross-linked treatment improved the viscoelastic property of hydrogels (Figure 5a and 5b). The storage moduli of CG_A and LCG_A after cross-linking with APTES are raised from 276 kPa and 580 kPa to 1110 kPa and 1391 kPa, respectively, indicating the addition of hydrolyzed APTES can cause mechanical enhancement. A 3-fold increase in storage modulus is noted after cross-linking, which confirms the formation of new cross-link bonding in the gel network.[47,64] Table 3 shows the comparison of mechanical properties of biobased hydrogels obtained from this study and those in the literature. The storage modulus (G') value of our APTES-reinforced lignocellulose hydrogel (LCG_A) is much higher than other biobased hydrogels. Strong mechanical property of hydrogels is required for the application of biobased hydrogel, for example, wastewater remedy by adsorption. Fan et al. reported that a mechanically strong graphene oxide/sodium alginate/polyacrylamide nanocomposite hydrogel was fabricated for improving the dye adsorption capacity.[17]

It should be noted that the improvement degree of mechanical behavior was influenced by addition of the silane coupling agent. Figure S5 shows the addition of 1% APTES is an optimal condition to achieve the highest storage modulus in the gel system. When the APTES content exceeded 1%, polysiloxane structures would form in the gel networks, which is attributed to formation of the −Si−O−Si− bond by self-condensation of silanols (Figure S1). The polysiloxane layer between the hydroxyl of LC molecules and the silanol of APTES results in weakening the strength of gels.

In addition, the storage modulus of the lignin-containing LCG is higher than that of CG, which increases from 276 to 580 kPa. The viscoelastic property of LCG_A is also better than CG_A. These analyzed results confirm the homodisperse lignin in the gel is responsible for the excellent strength properties.

Figure 5d shows the compressive strength of CG, LCG, CG_A, and LCG_A. As expected, the compressive strengths of gels after cross-linking with APTES have greatly improved. The compressive stress of CG_A and LCG_A at strains of 50% increased from 12 and 35 kPa to 55 and 78 kPa, and the corresponding compressive modulus in the elastic region (at

DOI: 10.1021/acssuschemeng.8b04413
ACS Sustainable Chem. Eng. 2019, 7, 2998–3009

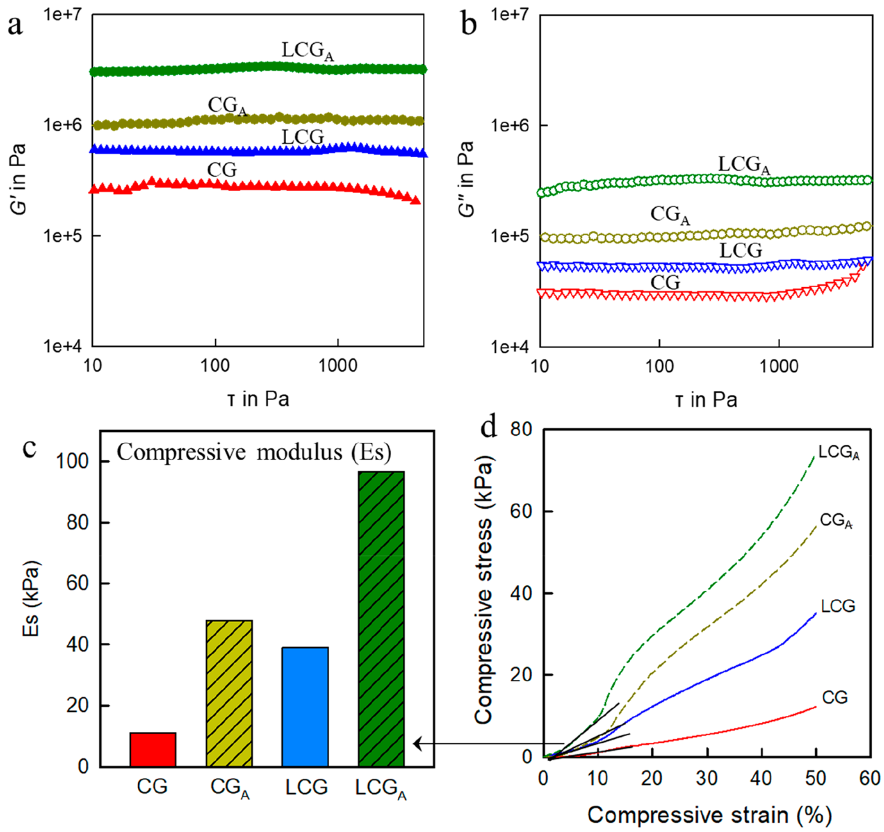


Figure 5. Mechanical properties of cellulose gel (CG), lignocellulose gel (LCG), and APTES-reinforced cellulose/lignocellulose gel (CG_A/LCG_A). (a) Storage modulus (G') and (b) loss modulus (G'') as a function of stress. (c) Compressive modulus (Es) of the gels at the elastic region. (d) Stress−strain curves with maximum strain of 50%.

Table 3. Comparison of Mechanical Properties of Biobased Hydrogels Obtained from This Study and Those in the Literature Work

	storage modulus G' (kPa)	compressive modulus Es (kPa)	ref
nanocellulose hydrogel	36.4	37.6	ref 65
CNC-PAAm hydrogel	7.50	--	ref 63
LCNF-PVA hydrogel	7.72	19.06	ref 10
catechol-Fe_3O_4 hydrogel	12	--	ref 66
chitin nanofiber hydrogel	8.35	--	ref 35
APTES-reinforced lignocellulose hydrogel (LCG_A)	1391	96	present work

10% strain) increases from 11 and 47 kPa to 46 and 97 kPa, respectively (Figure 5c). The improvement of compressive strength confirms the establishment of covalent linkages between the cellulose fibers and APTES, which ensures excellent mechanical strength in the hydrogels.[64]

Moreover, lignin-containing LCG and LCG_A samples exhibit better compressive properties than lignin-free CG and CG_A samples, indicating that the presence of lignin in the network plays a positive role in improvement of mechanical strength.

Antiultraviolet Aging Properties. Lignin, due to the existence of the aromatic ring structure, can absorb UV light, thus improving the anti-UV stability of biomaterial.[67] The weathering rate of materials is affected by exposure time, UV intensity, and irradiation distance. Under strong irradiation conditions (high-power UV intensity, long exposure time, and short irradiation distance), the organics can be decomposed to water and carbon dioxide.[68] The UV light transmittance of CG_A and LCG_A is shown in Figure 6a, and weight losses with increasing exposure time are shown in Figure 6b.

As shown in Figure 6a, the lignin-containing sample (LCG_A) possesses better UV light-shielding ability over the entire UV range, compared to the lignin-free sample (CG_A). When increasing the exposure time under the intense UV-light irradiation, the weight loss of the CG_A sample is significantly and consistently more, compared with that of the LCG_A sample (Figure 6b). For example, after 72 h of strong UV-light irradiation (high-power UV intensity and short irradiation distance), the remaining ratio of LCG_A is 90%, whereas the weight of CG_A remains only 70%. These results suggest that the presence of lignin plays a positive role in the UV-induced degradation process, which is ascribed to the antioxidation effect of lignin.[69] Fabiyi et al. investigated the effects of weathering on the constituents of the wood and polymer matrix behavior in HDPE-based wood plastic composites, and the results showed that the weight-average molecular weight (Mw) and number-average molecular weight (Mn) of extracted HDPE decreased with an increase in exposure time of the composite materials.[70] Thus, the results of Figure 6 indicate that the presence of lignin, as a UV quencher, plays a positive role in retarding the polymer degradation for the LCG_A sample.

Dye Adsorption Properties. Figure 7 shows the adsorption properties of prepared hydrogels for methylene blue (MB) and rhodamine B (RB) dyes with increasing contact time. The adsorption curves of hydrogels both present a trend of rapid increase and then reach equilibrium. The

DOI: 10.1021/acssuschemeng.8b04413
ACS Sustainable Chem. Eng. 2019, 7, 2998−3009

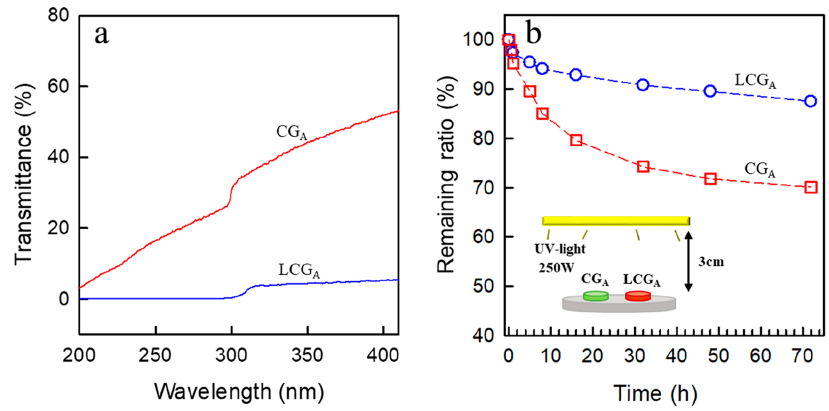


Figure 6. (a) UV light transmittance curves and (b) remaining ratio of lignin-free sample (CG$_A$) and lignin-containing sample (LCG$_A$) with the increase of exposure time under the UV-light irradiation with 250 W.

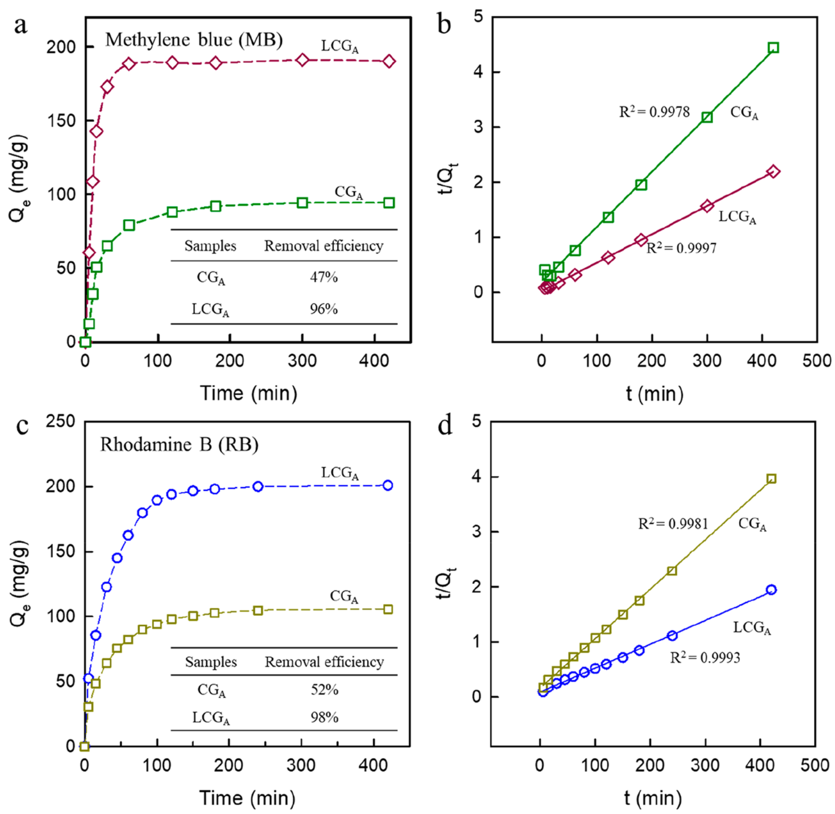


Figure 7. Dye adsorption capacity of lignin-free hydrogel (CG$_A$) and lignin-containing hydrogel (LCG$_A$) for methylene blue (MB) (a) and rhodamine B (RB) (c). The pseudo-second-order model of CG$_A$ and LCG$_A$ for MB (b) and RB (d) (organic dye: 50 mg L^{-1} (100 mL), adsorbent (o.d.) 25 mg).

adsorption rate of lignin-containing sample (LCG$_A$) is faster than the lignin-free sample (CG$_A$). The maximum equilibrium adsorption capacity of LCG$_A$ is 192 mg/g (MB) and 201 mg/g (RB), whereas that of CG$_A$ is 95 mg/g (MB) and 105 mg/g (RB). As shown in Figure 7a and 7c (inset), the adsorption efficiency of LCG$_A$ can reach 96% (MB) and 98% (RB), which is higher than that of CG$_A$ (MB: 47%, RB: 51%). The high adsorption capacities of lignocellulose hydrogels for organic dyes are attributed to two factors: First, organic dyes molecules can be adsorbed onto the surface of lignocellulose hydrogels by diffusion; then, organic dyes molecules further combine with active adsorption sites of the hydrogels. Table S1 shows the changes in zeta potential of CG$_A$ and LCG$_A$ before and after adsorbing MB, which is a cationic dye. The results showed that upon MB adsorption (67.89 mg/g CG$_A$ and 133.25 mg/g LCG$_A$) the zeta potential values of adsorbed MB-CG$_A$ and MB-LCG$_A$ changed from negative (due to the anionic nature of the cellulose/lignocellulose gels) to positive. These results indicate that the adsorption sites in the anionic hydrogel by the cationic MB dye are via the formation of electrostatic interactions, and furthermore, the excessive adsorbed MB is responsible for the positive charge in the zeta potential values after MB adsorption onto CG$_A$ and LCG$_A$ hydrogel samples. Compared with CG$_A$, LCG$_A$ exhibits better adsorption performance for MB dye, which is attributed to the polar functional groups of lignin[71,72] and large specific surface area.

DOI: 10.1021/acssuschemeng.8b04413
ACS Sustainable Chem. Eng. 2019, 7, 2998−3009

Table 4. Kinetic Parameters of the Pseudo-First-Order and Pseudo-Second-Order Kinetic Models for MB and RB Adsorption of the Lignin-Free Hydrogel (CG_A) and Lignin-Containing Hydrogel (LCG_A)

samples	$Q_{e,exp}$	pseudo-first-order kinetic			pseudo-first-order kinetic		
		R^2	$Q_{e,cal}$	K_1 (min^{-1})	R^2	$Q_{e,cal}$	K_2 (g/mg min)
Methylene Blue (MB)							
CG_A	94.5	0.962	63.81	0.0203	0.997	100.05	0.0005
LCG_A	191.9	0.947	35.42	0.0117	0.999	196.12	0.0009
Rhodamine B (RB)							
CG_A	105.7	0.996	72.66	0.0034	0.998	111.22	0.0003
LCG_A	200.9	0.984	139.47	0.0042	0.999	212.67	0.0005

Two kinetic models, the Lagergren pseudo-first-order and pseudo-second-order, were adapted to analyze adsorption kinetics of the lignocellulose hydrogels for organic dyes. The adsorption kinetic parameters are shown in Table 4. As shown, the R^2 from the pseudo-second-order model for hydrogels is higher than that of the pseudo-first-order model, and the theoretical Q_e values of the pseudo-second-order model are closer to the experimental results, which suggests that the adsorption kinetics of the as-prepared hydrogels fit well with the pseudo-second order model (Figure 7b and 7d). The adsorption behaviors of organic dyes onto the synthesized lignocellulose hydrogels are dominated by chemical sorption.[73] A similar effect was found by Uraki et al. that a lignin-containing hydrogel possesses higher adsorption capacity for organic dyes in comparison to the lignin-free hydrogel.[74] Thus, the developed lignocellulose gels are eco-friendly materials that will have promising applications for wastewater treatment.

Using lignocellulose hydrogels are advantageous over lignocellulose itself, for MB adsorption, because:

1) Hydrogels possess a three-dimensional network and porous structure.
2) Hydrogels have a large specific surface area, which can expose many active sites of functional groups benefiting for adsorption of organic dyes.
3) Hydrogels are easy to reuse/recycle after use.

Zhou et al. reported hydrolyzed polyacrylamide/cellulose nanocrystal (HPAM/CNC) nanocomposite hydrogels for MB dye removal, and the results showed that the synergy of CNCs and HPAM in a highly cross-linked networks is very effective for dye adsorption.[33] Yu et al. prepared lignosulfonate-*g*-acrylic acid (LS-*g*-AA) hydrogels for MB removal, verifying that as-prepared hydrogels are easy to reuse/recycle after the treatment.[34]

A comparison was carried out between the results of the adsorption capacities of APTES-reinforced lignocellulose hydrogel (LCG_A) and conventional adsorbents (such as activated carbons) or hydrogel-based adsorbents (such as cellulose-based hydrogel, lignin-based hydrogel, and tea waste-based composite) toward organic dyes. As shown in Table 5, the adsorption capacities obtained in this work are comparable with those reported in other literature even if different experimental conditions were used. This suggests that lignocellulose-based hydrogel can be used as an effective adsorbent for organic dye (such as MB, RB) removal. Furthermore, lignocellulose-based hydrogel is an environmentally friendly biosorbent with advantages of biocompatibility and biodegradability.

In summary, a novel mesoporous lignocellulose gel consisting of cellulose, lignin, and hemicellulose, with very high strength properties, was prepared. The enhanced mechanical properties of lignocellulose gels were achieved by using a silane-based coupling agent, which leads to the in situ formation of Si−O−C cross-links among the lignocellulosic fibrils. The addition of a small amount of silane-based coupling agent (e.g., 1 wt % APTES) is very effective: the dynamic storage modulus of APTES-reinforced lignocellulose gel (LCG_A) reaches up to 1391 kPa and the compressive modulus up to 96 kPa, a 3-fold increase in the dynamic storage modulus and a 2-fold increase in the compressive modulus, compared with the unmodified gel (LCG). The presence of lignin in the gel network also contributes to the improvement of mechanical strength. The storage modulus (580 kPa) of the lignin-containing LCG is higher than that of CG (276 kPa). Moreover, the as-prepared lignocellulose gels have other interesting properties, including antiultraviolet weathering and dye adsorption.

Table 5. Comparison of Adsorption Capacities of Biobased Hydrogels Obtained from This Study and Those in the Literature Work for Pollutant Removal

adsorbents	initial concentration of pollutants (mg/L)	adsorption capacity (mg/g)	removal efficiency (%)	ref
propylene diamine basic activated carbons	700 (MB)	182.0	--	ref 75
activated lignin-chitosan extruded blends	82 (MB)	36.25		ref 72
polyacrylamide/cellulose hydrogel	300 (MB)	326	90	ref 33
tea waste/dolomite composite	75 (MB)	150	89	ref 32
tannin-immobilized cellulose hydrogel	20 (MB)	1.1	91	ref 76
graphene−CNT hybrid aerogels	-- (RB)	145.9	--	ref 77
polydopamine-graphene hydrogel	600 (RB)	207.06	--	ref 78
APTES-reinforced lignocellulose hydrogel	50 (MB)	191.9	96	present work
	50 (RB)	200.9	98	present work

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssuschemeng.8b04413.

The zeta potential of CG_A and LCG_A before and after adsorbing MB. The chemical interaction of silane with lignocellulose molecules by hydrolysis process. FTIR spectra of APTES-reinforced lignocellulose. The con-

DOI: 10.1021/acssuschemeng.8b04413
ACS Sustainable Chem. Eng. 2019, 7, 2998−3009

focal image of pure cellulose gel (CG) labeled with AO. The 2D images of optical sectioning along the Z axis for the lignocellulose gel (LCG_A) labeled with AO. The storage modulus of CG under different additive amount of APTES with 0.5, 1.0, 1.5, and 2.0 wt % (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We are grateful for financial support from the National Key R&D Program of China (2017YFD0601005), as well as project funding from the National Natural Science Foundation of China (Grant No. 31870565), the Jiangsu Provincial Natural Science Foundation of China (BK20181397), the Doctorate Fellowship Foundation of Nanjing Forestry University, the Postgraduate Research and Practice Innovation Program of Jiangsu Province (KYCX17_0845), and the Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD).

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